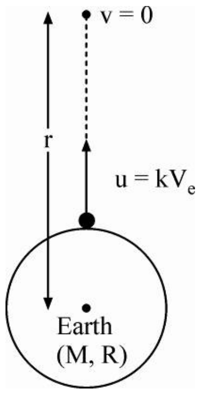
Solutions to JEE Advanced – 3 | JEE 2023 | Paper-1

PHYSICS

1.(C) Time depends on velocity component $\perp$ to river which is same for all
All the four boats A, B, C and D take same time.

2.(B) $u = kV_e = k\sqrt{\dfrac{2GM}{R}}$



$\Delta U + \Delta K = 0$

$$m\left(\frac{-GM}{r}\right) - m\left(\frac{-GM}{R}\right) + 0 - \frac{1}{2}mu^2 = 0$$

$$\Rightarrow \quad m\left(\frac{GM}{R}\right) - \frac{1}{2}mk^2\left(\frac{2GM}{R}\right) = \frac{mGM}{r}$$

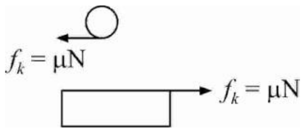
$$\Rightarrow \quad \frac{GMm}{R}(1-k^2) = \frac{GMm}{r} \quad \Rightarrow \quad r = \frac{R}{1-k^2}$$

$$\therefore \quad h = r - R = \frac{R}{1-k^2} - R = \frac{R\,k^2}{1-k^2}$$

3.(C) As the curve is smooth, there is no gain in angular motion.

$$\Rightarrow \quad Mgh = 1/2Mu^2 \quad \Rightarrow \quad u = \sqrt{2gh}$$

i.e. velocity with which it gets on to the plank.



$N\mu = MA$ [For plank]

$N = Mg$; $Mg\mu = MA$

$\Rightarrow \quad A = g\mu$ [Towards right]

[For cylinder]

$a = g\mu$ [Towards left]

$\tau_{cm} = I_{cm}\alpha$

$N\mu R = 1/2MR^2\alpha \quad \Rightarrow \quad \alpha = \dfrac{2g\mu}{R}$ (clockwise)

Let V and v be linear velocity of plank and center of cylinder when pure rolling begins

$V = g\mu t$

$v = u - g\mu t$

$\omega = \dfrac{2g\mu}{R}t$

When pure rolling begins

$$v - R\omega = V \quad \Rightarrow \quad (u - g\mu t) - 2g\mu t = g\mu t$$

$$\Rightarrow \quad t = \frac{u}{4g\mu}$$

$$\Rightarrow \quad v = \frac{3u}{4} \text{ and } V = \frac{u}{4}$$

$a_r = -2g\mu$; $u_r = u$

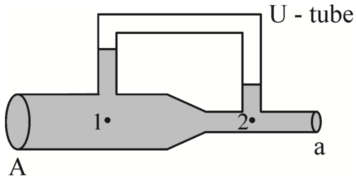
$$V_r = \frac{3u}{4} - \frac{u}{4} = \frac{u}{2} \quad \Rightarrow \quad \left(\frac{u}{2}\right)^2 = u^2 + 2(-2g\mu)\times l = \frac{3u^2}{16g\mu}$$

4.(C) Work $= \Delta KE = \frac{1}{2}mv^2 = \frac{1}{2}m(2a)(2\pi r) = 2\pi mra$

5.(A) Let velocity of water at the two cross-sections be v_1 and v_2

$\Rightarrow \quad Av_1 = av_2 = V$

Applying Bernoulli's Theorem between two points below the two arms of the U-tube at the centre of the pipe : [Let air pressure in U-tube = P_0]



$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

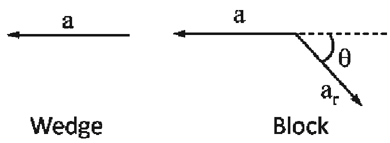
$$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

Using $\quad P_1 = P_0 + \rho g h_1, \quad P_2 = P_0 + \rho g h_2, \quad v_1 = \frac{V}{A}, \quad v_2 = \frac{V}{a}$

$$\rho g(h_1 - h_2) = \frac{1}{2}\rho\left(\frac{V^2}{a^2} - \frac{V^2}{A^2}\right) \quad \Rightarrow \quad h_1 - h_2 = \frac{V^2}{2g}\left(\frac{1}{a^2} - \frac{1}{A^2}\right)$$

6.(C) From constraint equation $a_r = 2a$

$\Rightarrow \quad$ Acceleration of block



$$= \sqrt{a^2 + (2a)^2 + 2(a)(2a)\,(\cos(\pi - \theta))}$$

$$= a\sqrt{5 - 4\cos\theta}$$

7.(BC) As air is pumped out, pressure above surface of water decreases because of which pressure at all points of water decreases $(P = P_{surface} + \rho g h)$.

So, force on bottom decreases.

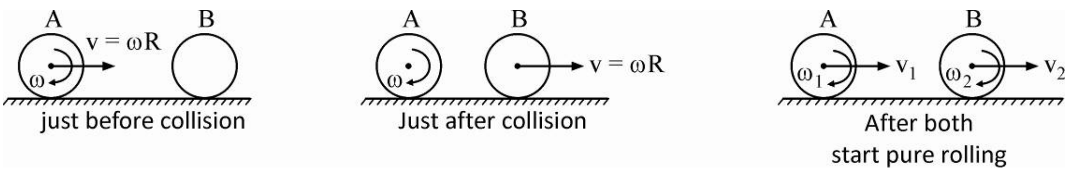
Since water is assumed to be incompressible, density and volume remain unchanged.

8.(BD) During collision, no impulsive friction force acts on A or B.

$\therefore \quad J_{CM} = 0 \Rightarrow \Delta L_{CM} = 0$

So, there is no change in angular velocity and just after collision $\omega_A = \omega$ & $\omega_B = 0$.

Their linear velocities get exchanged (same mass elastic head on collision).



After both start pure rolling :

For A $\quad : \quad \frac{2}{5}mR^2\omega = \frac{2}{5}mR^2\omega_1 + mV_1R$

$$\Rightarrow \quad \frac{2}{5}mR^2\omega = \frac{2}{5}mR^2\left(\frac{V_1}{R}\right) + mV_1R \Rightarrow V_1 = \frac{2}{7}\omega R$$

For B $\quad : \quad mVR = \frac{2}{5}mR^2\omega_2 + mV_2R$

$$\Rightarrow \quad m(\omega R)R = \frac{2}{5}mR^2\left(\frac{V_2}{R}\right) + mV_2R \Rightarrow V_2 = \frac{5}{7}\omega R$$

9.(BD) B cannot move in vertical line relative to A as horizontal component of their velocities cannot be equal.

$\vec{v}_{BA}$ is constant $\Rightarrow$ Straight line motion

Particle collide if $\vec{v}_{BA}$ is along BA

$\Rightarrow$ y-component of $\vec{v}_{BA} = 0$ $\Rightarrow$ $u_1 \sin\alpha = u_2 \sin\beta$

$\Rightarrow$ Time of flight is equal

If t is the time of collision and T is the time of flight, for collision : $t < T$

$$u_1 \cos\alpha t + u_2 \cos\beta t = d$$

$$u_1 \cos\alpha T + u_2 \cos\beta T = R_1 + R_2$$

$\because$ $T > t \Rightarrow R_1 + R_2 > d$

10.(ACD) At $x = 2R$, $F = -\dfrac{G(4M)m}{4R^2} + \dfrac{GMm}{16R^2} < 0$

$\Rightarrow$ Velocity is decreasing at $x = 2R$

For minimum velocity put $F(x) = 0$

$$\Rightarrow -\frac{G(4M)}{x^2} + \frac{GM}{(6R-x)^2} = 0 \Rightarrow x = 4R$$

So, minimum velocity occurs at $x = 4R$, and once this point is reached, the particle reaches surface of B.

Applying energy conservation : $K_f - K_i = U_i - U_f$

$$0 - \frac{1}{2}mv^2 = \left[-\frac{G(4M)m}{2R} - \frac{G(M)m}{4R}\right] - \left[-\frac{G(4M)m}{4R} - \frac{GMm}{2R}\right]$$

$$\Rightarrow v = \sqrt{\frac{3GM}{2R}}$$

11.(BCD)

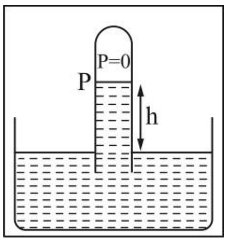
When stationary

$$\rho gh = P$$

When accelerating upwards

$$\rho(g+a)h_1 = P$$

$$\Rightarrow h_1 < h$$



When accelerating downwards

$$\rho(g-a)h_2 = P \Rightarrow h_2 > h$$

When moving with constant velocity

$$\rho gh_3 = P \Rightarrow h_3 = h$$

12.(ABD) After time t, mass of plank with sand will be $m_0 + \mu t$ and let its velocity be v. In next small time interval dt, let dm mass of sand falls with sped u and sticks to it.

Change in momentum of dm:

$$d\vec{p} = dm(v\hat{i}) - dm(-u\hat{j}) = dm(v\hat{i} + u\hat{j})$$

Force on $dm = \frac{d\vec{p}}{dt} = \frac{dm}{dt}\left(v\hat{i} + u\hat{j}\right) = \mu v\hat{i} + \mu u\,\hat{j}$

$\therefore$ Force on plank $= -\mu v\hat{i} - \mu u\,\hat{j} \quad \Rightarrow \quad F_{horizontal}\ \alpha\, v$ and $F_{vertical}\ \alpha\, u$

$\because \quad u = \sqrt{2gh} \qquad \therefore \qquad F_{\text{vertical}}\ \alpha\sqrt{h}$

Horizontal component of this force causes deceleration of plank.

$$\therefore \quad \left(m_0 + \mu t\right)\frac{dv}{dt} = -\mu v$$

$$\Rightarrow \quad \int_{V_0}^{V}\frac{-dv}{v} = \int_0^t \frac{\mu\, dt}{m_0 + \mu t} \quad \Rightarrow \quad \ell n\frac{v_0}{v} = ln\frac{m_0 + \mu t}{m_0}$$

$$\therefore \quad \frac{v_0}{v} = \frac{m_0 + \mu t}{m_0} \quad \Rightarrow \quad v = \frac{m_0 v_0}{m_0 + \mu t}$$

Let time taken to cross be T.

$$\frac{dx}{dt} = \frac{m_0 v_0}{m_0 + \mu t} \quad \Rightarrow \quad \int_0^{\ell} dx = m_0 v_0 \int_0^T \frac{dt}{m_0 + \mu t}$$

$$\Rightarrow \quad \ell = \frac{m_0 v_0}{\mu}\,\ell n\frac{m_0 + \mu T}{m_0} \quad \Rightarrow \quad m_0 + \mu T = m_0 e^{\frac{\ell\mu}{m_0 v_0}}$$

$\therefore$ Velocity after crossing

$$v = \frac{m_0 v_0}{m_0 + \mu T} = \frac{m_0 v_0}{m_0 e^{\frac{\ell\mu}{m_0 v_0}}} = v_0 e^{\frac{-\ell\mu}{m_0 v_0}}$$

$$a = \frac{dv}{dt} = \frac{-m_0 v_0 \mu}{\left(m_0 + \mu t\right)^2} = \frac{-\mu}{m_0 v_0}\left(\frac{m_0 v_0}{m_0 + \mu t}\right)^2 = \frac{-\mu}{m_0 v_0}v^2$$

1.(60) Minimum friction on plug $F = h\rho g \times \frac{\pi d^2}{4} = 5 \times 1000 \times 10 \times \frac{3 \times (0.04)^2}{4} = 60N$

2.(130) Ejection speed of water $= \sqrt{2gh}$

Volume flow rate $= \frac{\pi d^2}{4}\sqrt{2gh}$

Water flows in time t is $= \frac{\pi d^2}{4}\sqrt{2gh} \times t = \frac{3 \times (0.04)^2}{4}\sqrt{2 \times 10 \times 5} \times 3 \times 3600 \approx 130\ \text{m}^3$ (approx.)

3.(2) Time after which block will move

$$F\cos\theta \ge \mu_s N$$

$$F\cos\theta \ge \mu_s\left(mg - F\sin\theta\right)$$

$$F \ge \frac{\mu_s mg}{\left(\cos\theta + u_s\sin\theta\right)}$$

$$t \ge \frac{\mu_s mg}{\left(\cos\theta + \mu_s\sin\theta\right)\alpha}$$

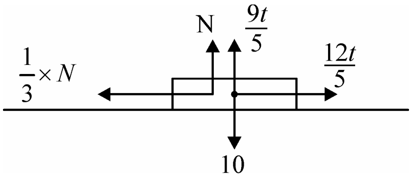
$$t_{\min} = \frac{\mu_s mg}{\alpha\sqrt{1 + \mu_s^2}} = 2\sec$$

When $\theta = 37°$

4.(28) For $\theta = 37°$, after $t = 2\,\text{sec}$

$$N = 10 - \frac{9t}{5}$$

Block leaves contact when $N = 0$



$\Rightarrow \quad 9t/5 = 10$

$$t = \frac{50}{9}s$$

Impulse $= \Delta\rho = mv$

$$\Rightarrow \quad v = \int_2^{50/9}\left(\frac{12t}{5} - \frac{N}{3}\right)dt = \int_2^{50/9}\left(3t - \frac{10}{3}\right)dt \approx 28\,m/s$$

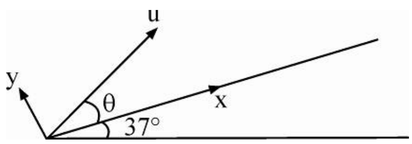
5.(1)

Taking x and y as shown, for motion of ball :

$$u_x = u\cos\theta + 5, u_y = u\sin\theta$$

$$a_x = -g\sin 37° = -6$$

$$a_y = -g\cos 37° = -8$$



At the centre at hoop, $Sy = 4$ and $V_y = 0$

$\therefore \quad V_y^2 = u_y^2 + 2a_y s_y \Rightarrow 0 = (u\sin\theta)^2 + 2(-8)(4)$

$\Rightarrow \quad u\sin\theta = 8$

Also $\quad V_y = u_y + a_y t \Rightarrow 0 = u\sin\theta + (-8)t \quad \Rightarrow \quad t = \dfrac{u\sin\theta}{8} = \dfrac{8}{8} = 1$

6.(10) $s = \sqrt{s_x^2 + s_y^2}$

$\Rightarrow \; s_x = \sqrt{s^2 - s_y^2} = \sqrt{80 - 16} = 8\,m$

$(u\cos\theta + 5)(1) + \dfrac{1}{2}(-6)(1) = 8 \Rightarrow u\cos\theta = 6$

$(u\sin\theta)^2 + (u\cos\theta)^2 = 8^2 + 6^2 = 100$

$\Rightarrow u = 10 \;\; m/s$

CHEMISTRY

1.(B) Nitrogen has half-filled configuration.

2.(A) Bond angle of $NH_3 >$ Bond angle of NF_3

3.(B) $K_P = P_{NH_3} \times P_{H_2S}$

$0.36 = P_{NH_3} \times P_{H_2S}$

$P_{NH_3} = P_{H_2S} = 0.6\,\text{atm}$

$P_{NH_3} + P_{H_2S} = 0.6 + 0.6 = 1.2\,\text{atm}$

4.(A) $\frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = R_H\left[\frac{3}{4}\right]$

5.(D) Moles of $HCl = M \times V = 5 \times 3.26 = 16.3$

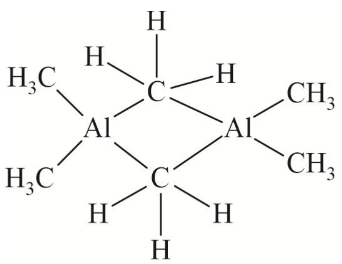
Moles of $MgCO_3 = \frac{16.3}{2} = 8.15$

Mass of $MgCO_3 = 8.15 \times 84 = 684.6\,g$

6.(A) $\text{I.E}_4 \text{ of Be} = \left(\frac{Z^2}{n^2} \times 13.6 \times 1.6 \times 10^{-19}\right)J = 16 \times 13.6 \times 1.6 \times 10^{-19} = 348.16 \times 10^{-19} = Y \times 10^{-19}\,J$

$\therefore \quad Y = 348.16$

7.(C) $Al(CH_3)_3$ is electron deficient, therefore it exist as dimer.



8.(CD) As temperature decreases or molar mass increases
H increases
L decreases
W decreases

9.(ABD) O_2^+ contains unpaired electron in π-antibonding M.O and N_2^+ in σ_{2p} orbital N_2^{2-} contains unpaired electron in π-antibonding M.O.

10.(CD) Weak acid and weak base gives less heat in neutralisation hence smaller temperature rise than in case of strong acids and bases if all other conditions are same.

11.(ACD) By definition.

12.(AC) **(A)** IE (I) of N is more than O due to stable half filled electronic configuration of valence shell in N.
(B) Electron gain enthalpy of O ($-141\ kJmol^{-1}$) is less than S, Se, Te and Po due to its small atomic size.
(C) Electronegativity on Mulliken scale is 2.8 times the electronegativity on Pauling scale.
(D) The ionic radius decreases as more electrons are ionized off.

1.(6)
$$\left.\begin{array}{ccc} 10 & \rightarrow & 4 \\ 9 & \rightarrow & 4 \\ 8 & \rightarrow & 4 \\ 7 & \rightarrow & 4 \\ 6 & \rightarrow & 4 \\ 5 & \rightarrow & 4 \end{array}\right\} \text{Total 6 spectral lines}$$

2.(8)

3.(525)

	$NH_4HS(s) \rightleftharpoons NH_3(g)$	$+ \quad H_2S(g)$
Initial partial pressure	P mm	0
Equilibrium partial pressure	$(P+x)$mm	x mm

From question, $P+x=625$ and $(P+x)+x=725$

$\therefore x=100$ and $P=525$

4.(250) $P_{NH_3} = P_{H_2S} = \sqrt{K_P} = 250$ mm of Hg

5.(576) $w=-nR\Delta T = -1\times 8\times 72 = -576$ J

6.(1.56) $\Delta U = q + w$

$=1.6-0.576 = 1.024$ kJ

$$\gamma = \frac{\Delta H}{\Delta U} = \frac{1.6}{1.024} = 1.56$$

MATHEMATICS

1.(B) All the possible number are ${}^9C_5$ (none containing the digit 0)

$=126$.

Total numbers starting with

$1={}^8C_4=70$

1				

(using $2,3,4,5,6,7,8,9$) Total numbers starting with $23={}^6C_3=20$

2	3			

(using $4,5,6,7,8,9$) Total numbers starting with $245={}^4C_2=6$

2	4	5		

(using $6,7,8,9$)

97^{th} number =

2	4	6	7	8

2.(D) Let roots of $x^3-ax^2+bx-c=0$ be α,β,γ

$\Rightarrow \quad a=\alpha+\beta+\gamma,\ b=\alpha\beta+\beta\gamma+\gamma\alpha, c=\alpha\beta\gamma$

Now, $(\alpha+1),(\beta+1),(\gamma+1)$ are the roots of $x^3+px^2+qx-9=0$

$\Rightarrow \quad (\alpha+1)(\beta+1)(\gamma+1)=9 \Rightarrow \quad \alpha\beta\gamma+\alpha+\beta+\gamma+\alpha\beta+\beta\gamma+\gamma\alpha+1=9 \Rightarrow a+b+c=8$

3.(C) $$\frac{\sum_{r=0}^{n} nC_r \sin 2rx}{\sum_{r=0}^{n} nC_r \cos 2rx}=\frac{\sum_{r=0}^{n} nC_{n-r}\sin 2(n-r)x}{\sum_{r=0}^{n} nC_{n-r}\cos 2(n-r)x}=\frac{\sum_{r=0}^{n} nC_r\left[\sin 2rx+\sin 2(n-r)x\right]}{\sum_{r=0}^{n} nC_r\left[\cos 2rx+\cos 2(n-r)x\right]}$$

$$=\frac{\sum\limits_{r=0}^{n} {}^nC_r\, 2\sin nx\cos(2r-n)x}{\sum\limits_{r=0}^{n} {}^nC_r\, 2\cos nx\cos(2r-n)x}=\tan nx.$$

4.(D) $t_{r+1}=rS_r=r\left[t_r+S_{r-1}\right]=r^2S_{r-1}=r^2(r-1)S_{r-2}$

$\Rightarrow t_{r+1}=r(r!)t_1 \qquad (\because S_1=t_1)$

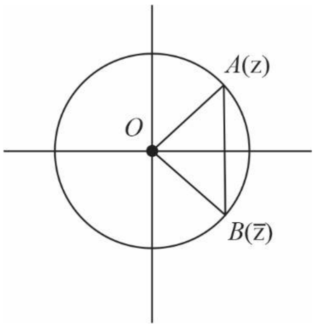
$\Rightarrow S_r=(r!)t_1$

$t_{r+1}+S_r=(r+1)S_{r+1}-(r+1)t_{r+1}$

5.(D) $|z-\overline{z}|=$ straight line AB

while $|z|(\arg z-\arg\overline{z})=Arc\ AB$

$\therefore\ |z-\overline{z}|\le|z|(\arg z-\arg\overline{z})$



6.(A) We have $\sin x\cos y=a^2$ ......(1)

and $\cos x\sin y=a$ ......(2)

Adding the equations (1) and (2), we get $\sin x\cos y+\cos x\sin y=a^2+a$

$\sin(x+y)=a^2+a \Rightarrow -1\le\sin(x+y)\le1\Rightarrow-1\le a^2+a\le1$

$a^2+a+1\ge0$ and $a^2+a-1\le0$

$a^2+a+1\ge0$ is true for all values of $a\in R$ and $a^2+a-1\le0\Rightarrow a\in\left[-\frac{\sqrt{5}+1}{2},\ \frac{\sqrt{5}-1}{2}\right]$ ......(3)

And subtracting equations (1) and (2)

$\sin x\cos y-\cos x\sin y=a^2-a$; $\sin(x-y)=a^2-a$; $-1\le a^2-a\le1$

$a\in\left[\frac{1-\sqrt{5}}{2},\ \frac{1+\sqrt{5}}{2}\right]$ ......(4)

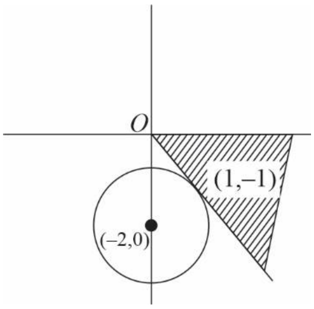
Taking common solution of equation (3) and (4).

$a\in\left[\frac{1-\sqrt{5}}{2},\ \frac{\sqrt{5}-1}{2}\right]$.

7.(BC) $\frac{kz}{k+1}$ represent any point lying on the line joining origin and z. So, $\frac{kz}{k+1}$ should lie outside the circle $|z+2i|>\sqrt{2}$.

So, z should lie in the shaded region

$\Rightarrow\quad -\frac{\pi}{4}<\arg(z)<0.$



8.(ABC)

$\sin^2 x - a\sin x + b = 0$ has only one solution in $(0, \pi)$

$\Rightarrow$ $\sin x = 1$ gives one solution and $\sin x = \alpha$ gives other solution such that $\alpha > 1$ or $\alpha \le 0$

$\Rightarrow$ $(\sin x - 1)(\sin x - \alpha)$ is the same equation as $\sin^2 x - a\sin x + b = 0$

$\Rightarrow$ $1 + \alpha = a$ and $\alpha = b$

$\Rightarrow$ $1 + b = a$ and $b > 1$ or $b \le 0$

$\Rightarrow$ $b \in (-\infty, 0] \cup [1, \infty)$ and $a \in (-\infty, 1] \cup [2, \infty)$.

9.(AD)

10.(AB)

x cannot be odd integer for if x is odd, x^2 is odd but $2px + 2q$ is even;

so $x^2 + 2px + 2q \ne 0$

x cannot be even integer for if x is even, $x^2 + 2px$ is a multiple of 4 but $2q$ is not.

So $x^2 + 2px + 2q \ne 0$

Also $(x + p)^2 = p^2 - 2q$

$\Rightarrow$ If x is fraction then $(x + p)^2$ is also a fraction but $p^2 - 2q$ is an integer. So, roots cannot be integer or rational numbers.

11.(ABC)

12.(BD)

The three possible discriminants are

$c^2 - 4ab,\ b^2 - 4ac,\ a^2 - 4bc$

For $a + b + c = 0$ they are perfect square.

1.(5) For the expansion $(1 + x)^n = \sum_{r=0}^{n} {}^nC_r x^r$, we have

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots + {}^nC_n = 2^n \qquad \ldots(i)$$

$$(1+\omega)^n = {}^nC_0 + {}^nC_1\omega + {}^nC_2\omega^2 + {}^nC_3 + {}^nC_4\omega + {}^nC_5\omega^2 + {}^nC_6 + \ldots\ldots., \qquad \ldots(2)$$

$$\left(1+\omega^2\right)^n = {}^nC_0 + {}^nC_1\omega^2 + {}^nC_2\omega + {}^nC_3 + {}^nC_4\omega^2 + {}^nC_5\omega + {}^nC_6 + \ldots\ldots \qquad \ldots(3)$$

Add (1), (2) and (3), we get

$$2^n + \left(-\omega^2\right)^n + (-\omega)^n = 3\left[\ {}^nC_0 + {}^nC_3 + {}^nC_6 + \ldots\ldots..\right]$$

$$\left(\because\ \omega^3 = 1 \text{ and } 1+\omega+\omega^2 = 0\right)$$

$$\therefore \qquad {}^nC_0 + {}^nC_3 + {}^nC_6 + \ldots. = \frac{1}{3}\left[2^n + (-1)^n\left(\omega^n + \omega^{-n}\right)\right] \qquad \left(\because\ \omega^2 = \overline{\omega}\right)$$

$$= \frac{1}{3}\left[2^n + (-1)^n.2\cos\frac{2n\pi}{3}\right] \quad \left[\because\ \omega = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right]$$

$$= \frac{1}{3}\left[2^n + (-1)^n.2\cos\left(n\pi - \frac{n\pi}{3}\right)\right]$$

$$= \frac{1}{3}\left[2^n + 2(-1)^n(-1)^n\cos\frac{n\pi}{3}\right] = \frac{1}{3}\left[2^n + 2\cos\frac{n\pi}{3}\right] \qquad \Rightarrow \qquad \lambda = 2;\ \mu = 3$$

2.(0) Let $n = 2\ m$ where m is positive integer then $k = 3m$.

We have

$$(1+x)^{6m} = 1 + {}^{6m}C_1x + {}^{6m}C_2x^2 + {}^{6m}C_3x^3 + {}^{6m}C_4x^4 \ldots\ldots + {}^{6m}C_{6m}x^{6m}$$

Put $x = \sqrt{3}i$, we get

$$\left(1+\sqrt{3}i\right)^{6m} = 1 + {}^{6m}C_1\sqrt{3}i - {}^{6m}C_2 3. - {}^{6m}C_3 3\sqrt{3}i + {}^{6m}C_4 9 + \ldots. + {}^{6m}C_{6m}\left(\sqrt{3}i\right)^{6m}$$

$$\Rightarrow \qquad 2^{6m}\left[\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right]^{6m} = \left(1 - {}^{6m}C_2.3 + {}^{6m}C_4.3^2 - \ldots\ldots...\right) + \sqrt{3}i\left({}^{6m}C_1 - {}^{6m}C_3 3 + \ldots.\right)$$

Equating imaginary parts we get

$$\sqrt{3}\left({}^{6m}C_1 - {}^{6m}C_2.3 + {}^{6m}C_2.3^2 - \ldots\ldots\right) = 2^{6m}\sin 2m\pi = 0 \qquad \therefore \qquad \sum_{r=1}^{3m}(-3)^{r-1}\ {}^{6m}C_{2r-1} = 0$$

3.(9) $V_r = \frac{r}{2}\left[2(r) + (r-1)(2r-1)\right] = \frac{1}{2}\left(2r^3 - r^2 + r\right)$

$$\therefore \qquad \sum_{k=1}^{n} V_k = \sum_{k=1}^{n} k^3 - \frac{1}{2}\sum_{k=1}^{n} k^2 + \frac{1}{2}\sum_{k=1}^{n} k$$

$$= \frac{1}{4}n^2(n+1)^2 - \frac{1}{12}n(n+1)(2n+1) + \frac{1}{4}n(n+1)$$

$$= \frac{1}{12}n(n+1)\left[3n(n+1) - (2n+1) + 3\right]$$

$$=\frac{1}{12}n(n+1)\left(3n^2+n+2\right)\Rightarrow a=12, b=3, c=2$$

$$T_r=V_{r+1}-V_r-2$$

$$=\left[(r+1)^3-r^3\right]-\frac{1}{2}\left[(r+1)^2-r^2\right]+\frac{1}{2}\left[(r+1)-r\right]-2$$

$$=\left(3r^2+3r+1\right)-\frac{1}{2}(2r+1)+\frac{1}{2}-2$$

$$=3r^2+2r-1=(3r-1)(r+1)\Rightarrow d=3, e=1$$

$$\therefore a+b-c-d-e=12+3-2-3-1=9$$

4.(6) $W_r=\frac{1}{T_r}+\frac{1}{4(r+1)}=\frac{1}{(3r-1)(r+1)}+\frac{1}{4(r+1)}$

$$=\frac{3}{4(3r-1)}\Rightarrow p=3, q=3\Rightarrow p-q=0$$

$$Q_r=T_{r+1}-T_r=3\left\{(r+1)^2-r^2\right\}+2\left\{(r+1)-r\right\}=3(2r+1)+2=5+6r$$

$\therefore$ $Q_1, Q_2, ...$ forms an A.P. with common difference 6.

$\Rightarrow$ $\lambda=6$

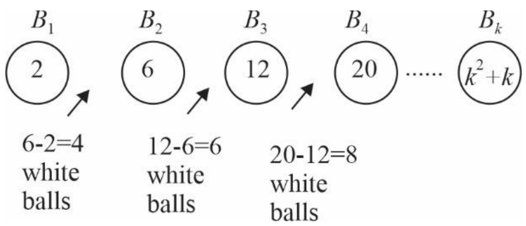
5.(18) Let $B_1, B_2, B_3, ..., B_k$ black balls are placed in a row.

Printed number on B_1 is 1^2+1 i.e. 2

Printed number on B_2 is 2^2+2 i.e. 6

$\vdots$

Printed number on B_k is k^2+k.



Similarly the number of white balls between last two black balls $=\left(k^2+k\right)-\left[(k-1)^2+(k-1)\right]=2k$

$\therefore$ Total number of white balls $=4+6+8+....+2k$

$=k^2+k-2$ ...(i)

Since, number of white balls $=340$

$\therefore (i)\Rightarrow k^2+k-2=340\Rightarrow k=18$

6.(9) Number of white balls available $=140$

Number of black balls available $=12$

If $k=12;$ then $k^2+k-2=154>140$

If $k=11;$ then $k^2+k-2=130$

Hence, $\lambda=140-130=10$ and $\mu=12-11=1$

$\Rightarrow \quad \lambda-\mu=9$

Solutions to JEE Advanced – 3 |JEE 2023 | Paper – 2

PHYSICS

1.(ABC) **(A)** Put $t=\frac{1}{l}$ $x=\frac{k}{l}\left[1-\frac{1}{e}\right]=\frac{k}{l}\left[1-\frac{1}{2.71}\right]=\frac{k}{l}\left[\frac{1.71}{2.71}\right]\approx\frac{k}{l}\frac{2}{3}$

(B) $\frac{dx}{dt}=v=ke^{-lt}$, $\frac{d^2x}{dt^2}=a=-lke^{-lt}$ put $t=0$

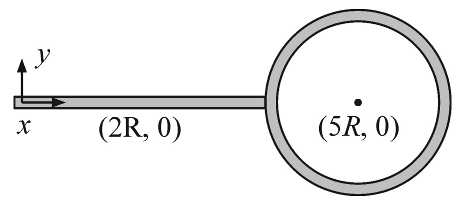
(C) Particle starts from $x = 0$ and maximum displacement achieved is at $t\to\infty$ and $x\to\frac{k}{l}$

$\therefore$ Particle cannot go beyond $x=\frac{k}{l}$

(D) Wrong

2.(BC) Location of $c.m.$ is given by $y_{cm}=0,\ x_{cm}=\frac{m(2R)+2m(5R)}{m+2m}=4R$

$\therefore$ $c.m.$ is at the junction



$I_{cm}=I_{rod}+I_{ring}$

$=\frac{m(4R)^2}{3}+2(2m)R^2=\frac{28}{3}mR^2$

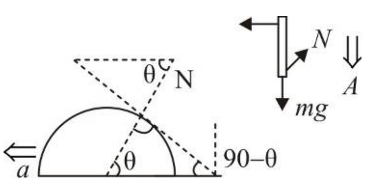
$I_z=I_{cm}+(3m)\ (4R)^2=\frac{28}{3}mR^2+48mR^2=\frac{172}{3}mR^2$

3.(AD) The progression of velocities of A, B and C will be

$3v,\ 2v,\ v \quad\Rightarrow\quad 2v,\ 3v,\ v \quad\Rightarrow\quad 2v,\ v,\ 3v \quad\Rightarrow\quad v,\ 2v,\ 3v$

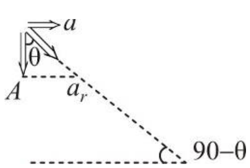
Number of collisions $= 3$

4.(AD)



$N\cos\theta=Ma$

$mg-N\sin\theta=mA$

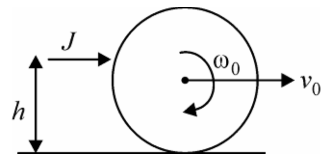


$\tan\theta=\frac{a}{A}\quad\Rightarrow\quad A=a\cot\theta$

3 equations, 3 unknowns.

5.(ACD) $J = mv_0$ ....(i)

$J \times (h - R) = \frac{mR^2\omega_0}{2}$ ....(ii)

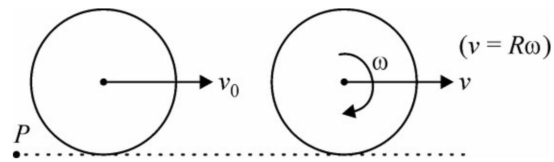


If $h = \frac{3R}{2} \Rightarrow \omega_0 = \frac{v_0}{R}$ (Disc starts Rolling)

If $h = R$

$v_0 = \frac{J}{m}$, $\omega_0 = 0$

Disc will start rolling after some time



About P : $L = mv_0R = mvR + \frac{mR^2}{2}\frac{v}{R}$

$v_0 = \frac{3v}{2} \Rightarrow v = \frac{2v_0}{3} = \frac{2J}{3m}$

In the frame of center of mass bottommost point is moving backward & friction is also acting backward so $W_{fric} > 0$

6.(AD) Pressure at base $= P = \rho g h$ $\Rightarrow$ Force on base $= P \times \pi a^2$ $= \rho\pi g h a^2$

Note that this force is exactly equal to the weight of cylindrical volume of water directly above the base [The cylinder of radius a and height h]

The weight of the rest of the water is supported by the curved surface.

Total weight of water in bucket

$W = \rho$ [Volume of frustum, V]g.

$V = \frac{\pi h}{3}[a^2 + ab + b^2]$

Net force on curved surface $= W - \rho\pi g h a^2$

Net force on bucket $= W$.

7.(D) Conserve angular momentum

$$\frac{1}{2}MR^2\omega = \left[\frac{1}{2}MR^2 + \frac{1}{2}\left(\frac{M}{8}\right)\left(\frac{R^2}{4}\right)\right]\omega' \qquad \Rightarrow \qquad \omega' = \frac{32\omega}{33}$$

8.(C) Consider smaller disc, take radius = r and mass = m

Take ring shaped element of radius x and width dx.

Torque due to friction on this ring element = $d\tau = \frac{\mu mg(2\pi x dx)x}{\pi r^2}$

Total torque on smaller disc = $\tau = \int_0^r d\tau = \frac{2\mu mg}{r^2}\int_0^r x^2\, dx = \frac{2}{3}\mu mgr$

$\tau = I\alpha \quad \Rightarrow \frac{2}{3}\mu mgr = \frac{mr^2}{2}\alpha \quad \Rightarrow \quad \alpha = \frac{4\mu g}{3r}$

$0 + \alpha t = \omega' \quad \Rightarrow \quad \frac{4\mu g}{3r}t = \frac{32}{33}\omega \quad \Rightarrow \quad t = \frac{8\omega r}{11\mu g} = \frac{4\omega R}{11\mu g}$

9.(B) Let total volume be V.
Just after release, velocity is zero, so viscous drag force is zero

$$mg - B = ma \Rightarrow V\sigma g - V\rho g = V\sigma a \Rightarrow a = \left(1 - \frac{\rho}{\sigma}\right)g.$$

10.(A) For the system of two spheres :

$$m_1 g + m_2 g - B_1 - B_2 = Fd_1 + Fd_2$$

$$\frac{4}{3}\pi(8r^3 + r^3)\sigma g - \frac{4}{3}\pi(8r^3 + r^3)\rho g = 6\pi\eta(2r + r)V_T$$

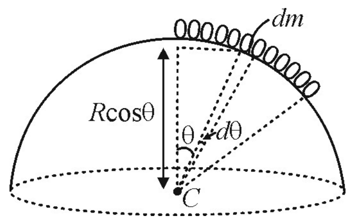
$$\Rightarrow \quad \frac{4}{3}\pi(9r^3)g(\sigma - \rho) = 6\pi\eta(3r)V_T \quad \Rightarrow \quad V_T = \frac{2}{3}\frac{(\sigma-\rho)gr^2}{\eta}$$

11.(B) GPE of elemental mass 'dm' is :

$$dU = (dm)gR\cos\theta$$

$$dU = \left(\frac{m}{\ell}Rd\theta\right)gR\cos\theta$$

$$dU = \frac{mgR^2}{\ell}\cos\theta d\theta$$



$\therefore$ Total GPE is : $U = \int dU$

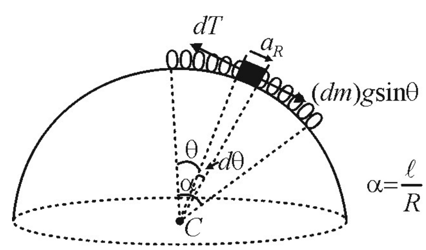
$$\Rightarrow \quad U = \int_0^{\ell/R}\frac{mgR^2}{\ell}\cos\theta d\theta$$

$$\Rightarrow \quad U = \frac{mgR^2}{\ell}\sin\left(\frac{\ell}{R}\right)$$ i.e. option (B)

12.(D) Consider an elemental length $Rd\theta$ at an angle θ with the vertical.
Using $F = ma \Rightarrow (dm)g\sin\theta - dT = (dm)a_R$

Integrating both sides,

$$\int(dm)g\sin\theta - \int dT = \int dm\, a_R; \quad \frac{dm}{Rd\theta} = \frac{m}{\ell}$$

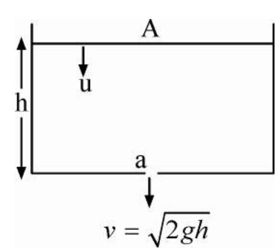


$$\Rightarrow \quad \int_0^{\alpha}\frac{m}{\ell}gR\sin\theta d\theta - \int_0^0 dT = \int_0^{\alpha}\frac{m}{\ell}a_R R d\theta; \quad a_R = \text{Constant}$$

$$\Rightarrow \quad \frac{mgR}{\ell}[1 - \cos\alpha] = \frac{m}{\ell}a_R R[\alpha]$$

$$\Rightarrow \quad \frac{gR}{\ell}[1 - \cos\alpha] = a_R \Rightarrow a_R = \frac{gR}{\ell}\left[1 - \cos\frac{\ell}{R}\right]$$ i.e. option (D)

1.(2) $Au = a\sqrt{2gh}$



$$\Rightarrow \quad A\left(-\frac{dh}{dt}\right) = a\sqrt{2gh} \quad \Rightarrow \quad \int_{H_1}^{H_2} -\frac{dh}{\sqrt{h}} = \int_0^t \frac{a}{A}\sqrt{2g}\, dt$$

$$\Rightarrow \quad -\left[2\sqrt{h}\right]_{H_1}^{H_2} = \frac{a}{A}\sqrt{2g}t \quad \Rightarrow \quad t = \frac{A}{a}\sqrt{\frac{2}{g}}[\sqrt{H_1} - \sqrt{H_2}]$$

Now, $T_1 = \frac{A}{a}\sqrt{\frac{2}{g}}\left[\sqrt{H} - \sqrt{\frac{H}{9}}\right]$ and $T_2 = \frac{A}{a}\sqrt{\frac{2}{g}}\left[\sqrt{\frac{H}{9}} - \sqrt{0}\right]$ $\quad\therefore\quad$ $\frac{T_1}{T_2} = \frac{2}{3}\sqrt{H} / \frac{1}{3}\sqrt{H} = 2$

2.(4) In equilibrium : $F_1 + F_2 = mg$ . . . . . (i)

After upper spring is cut F_1 becomes zero but F_2 remains unchanged

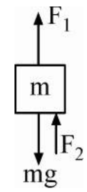
$\therefore \quad mg - F_2 = m(6)$ . . . . .(ii)

If lower spring is cut, F_2 becomes zero but F_1 remains unchanged

$\therefore \quad mg - F_1 = m(a)$

From (i) and (ii), $F_1 = mg - F_2 = 6m \quad \therefore \quad mg - 6m = ma$

$a = g - 6 = 4 m/s^2$



3.(5) At instant of projection, $v_{cm} = \dfrac{m(0) + m(+u)}{2m} = \dfrac{u}{2}$

For CM, $\quad v^2 = \left(\dfrac{u}{2}\right)^2 + 2(-g)\left(-\dfrac{h}{2}\right) = 3^2 + 2 \times 10 \times 0.8 \quad \Rightarrow \quad v = 5$

4.(2) In orbit particle's $PE = -mv^2$

To take it to infinity it has to reach infinity with at least $TE = 0$

So KE in orbit + PE in orbit should be zero.

$\therefore$ KE in orbit should be $mv^2 \quad \therefore \quad n = 2.$

5.(5) Let ρ_1 = density of water, ρ_2 = density of oil and σ = surface tension of water.

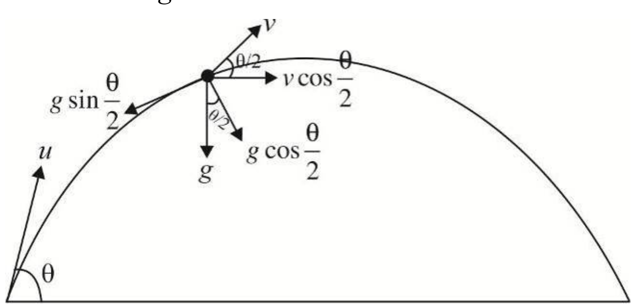
Equating pressure at the bottom of the capillary.

$$P_{atm} - \frac{2\sigma}{r} + \rho_1 g h_2 = P_{atm} + \rho_2 g h_2 \quad \Rightarrow \quad r = \frac{2\sigma}{(\rho_1 - \rho_2) g h_2}$$

Solve to get $r = 5 \times 10^{-4}\, m \quad \Rightarrow \quad \beta = 5$

6.(2) $v \cos\dfrac{\theta}{2} = u \cos\theta \;\Rightarrow\; v = \sqrt{10}\, m/s.$

Using $a_n = \dfrac{v^2}{R}$, $R = \dfrac{10}{g \cos 30^\circ} \;\Rightarrow\; \sqrt{3}R = 2$



CHEMISTRY

1.(ACD)

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

$$CO_3^{2-} + HCl \longrightarrow HCO_3^- + Cl^-$$

Now, methyl orange indicator is added and following reaction occurred further:

$$HCO_3^- + HCl \longrightarrow H_2O + CO_2 + Cl^-$$

Let solution has x mmol NaOH and y mmol CO_3^{2-}

$\Rightarrow$ y = 15 and x + y = 35

$\Rightarrow$ x = 20

$\Rightarrow$ Total mmol of carbonate in 100 mL solution = 15

Total mmol of NaOH in solution = 20 = 0.80 g NaOH)

Mass % of $M_2(CO_3)_3 = \frac{1.2}{2} \times 100 = 60\%$

$\Rightarrow$ Molar mass of $M_2(CO_3)_3 = \frac{1.2}{5 \times 10^{-3}} = 240 \quad \Rightarrow \quad 240 = 2 \times M + 60 \times 3$

And atomic weight of metal (M) = 30

2.(CD) Based on Le-Chatelier's principle.

3.(ABC) PCl_5 has trigonal pyramidal structure.

4 (ABCD) Fact

5.(BC) Since intermolecular forces are zero hence potential energy increases with decreasing distance, while Z increases by increasing pressure.

6.(ABD)

$$\Delta G^0 = \Delta H^0 - T\Delta S^0$$

Also $\Delta G^0 = -RT \ln K_p$

$\Rightarrow$ $-RT \ln K_p = \Delta H^0 - T\Delta S^0$

$$RT \ln K_P = T\Delta S^0 - \Delta H^0$$

$\ln K_p = \frac{1}{R}\left[\Delta S^0 - \frac{\Delta H^0}{T}\right]$; Hence (A), (B), (D) is correct

7.(C) $D = \frac{\text{mol. mass of } SO_3}{2} = \frac{80}{2} = 40$

8.(C) $\alpha = \frac{40 - 30}{(1.5 - 1)30} = \frac{10}{15} = 0.66$

9.(B) As ΔS is a state function.

10.(A) A to C having constant volume therefore this process is isochoric
C to B having constant pressure therefore this process is isobaric.

11.(B), 12.(B)

$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$ 0.025 mole $\qquad \dfrac{1\times1.23}{0.082\times300}=0.05$ = 0.025 × 63.5 = 1.5875g $\therefore$ Percentage of $Cu=\dfrac{1.5875}{1.9145}\times100=82.92\%$ and Zn present = (1.9145 – 1.5875) = 0.327g $=\dfrac{0.327}{65.4}=0.005\,\text{mole}$	$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$ 0.025 mole 0.1 mole $4Zn+10HNO_3 \rightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$ 0.005 mole 0.0125 mole $\qquad$ 0.00125 × 80 = 0.1g Now, moles of $HNO_3, \dfrac{V\times3}{1000}=0.1125$ $\therefore$ V = 37.5 mL

1.(344.80)

Work is done against constant pressure and thus irreversible.

$\Delta V = 4\,\text{lit}$

P = 1 atm

$w=-1\times4\,\text{lit-atm}=-1\times4\times101.3=-405.2\,\text{Joule}$

$\Rightarrow \Delta E = q + w$

$\therefore \Delta E = 750 - 405.2$

$\Delta E = +344.80\,J$

2.(30.60)

$$KE = 13.6\times\frac{Z^2}{n^2}=13.6\times\frac{3^2}{2^2}=30.6\,eV$$

3.(2763.60)

$\Delta G^\circ = -2.303RT\log K_P$

$= -2.303\times2\times300\times\log10^{-2} = 2763.60$ cal

4.(353.25)

$u_{av} = u_{rms}$

$$\sqrt{\frac{8RT_1}{\pi M}}=\sqrt{\frac{3RT_2}{M}}$$

$$\frac{8RT_1}{\pi M}=\frac{3RT_2}{M} \text{ or } \frac{8RT_1}{\pi M}=\frac{3R\times300}{M}$$

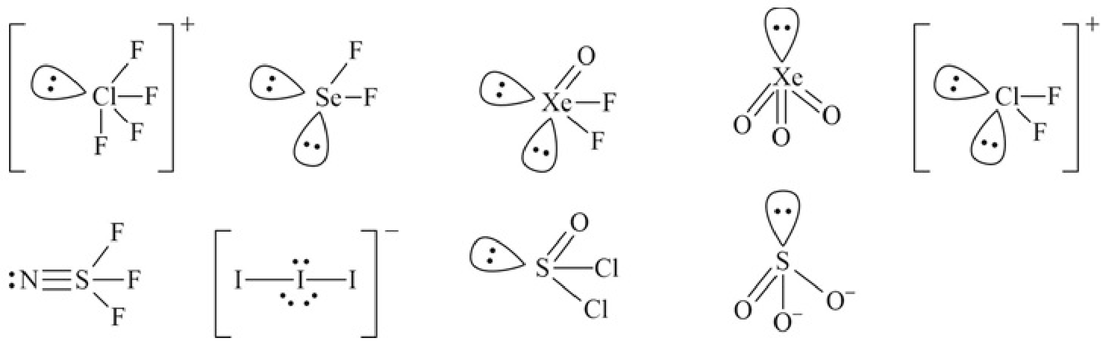
$T_1 = 353.25\,K$

5.(80.32)

$$\Delta H = \Delta E + \Delta n_g RT$$

$$\Rightarrow \quad 82.8 = \Delta U + \frac{1 \times 8.314 \times 298}{1000} \quad \Rightarrow \quad \Delta U = 80.32$$

6.(3)



MATHEMATICS

1.(BC)

At $x = 0$, equation is satisfying.

$$\left(2^x - 1\right)x^2 + \left(2^{x^2-1} - 1\right)x = \left(2^x - 1\right)$$

Comparing coefficients

$$x^2 = 1, \ \left(2^{x^2-1} - 1\right) = 0, \ \& \, x = 1$$

$$\Rightarrow \quad x = \pm 1$$

$$\therefore \quad x = -1, 0, 1$$

2.(AC) We have $\left(1 - 2x + 3x^2 - 4x^3 + ...\right)^{-n}$

$$= \left[\left(1 + x\right)^{-2}\right]^{-n} = \left(1 + x\right)^{2n}$$

So, coefficient of x^n in $\left(1 - 2x + 3x^2 - 4x^3 + ...\right)^{-n}$

$$= \text{coefficient of } x^n \text{ in } \left(1 + x\right)^{2n} = {}^{2n}C_n = \frac{(2n)!}{(n!)^2}.$$

3.(BC) We have $\left(x^2+\frac{1}{x^2}-2\right)^{10} \Rightarrow \left(x-\frac{1}{x}\right)^{20}$

$T_{r+1}={}^{20}C_r\,(x)^{20-r}(-x)^{-r}$

$T_{r+1}={}^{20}C_r\,(x)^{20-2r}(-1)^r$

The value of constant term in the binomial expansion $\left(x-\frac{1}{x}\right)^{10}$, if

$20-2r=0 \Rightarrow r=10$

$T_{10+1}={}^{20}C_{10}(-1)^{10}={}^{20}C_{10}$

(A) Number of different dissimilar term in the expansion

$(x_1+x_2+x_3+.....+x_{10})^{10}={}^{10+10-1}C_{10-1}={}^{19}C_9$

(B) $\left({}^{10}C_0\right)^2+\left({}^{10}C_1\right)^2+......+\left({}^{10}C_{10}\right)^2=\frac{(2\times10)!}{10!10!}=\frac{20!}{10!10!}={}^{20}C_{10}$

$\left[\because \left({}^{n}C_1\right)^2+\left({}^{n}C_2\right)^2+......+\left({}^{n}C_n\right)^2=\frac{2n!}{n!n!}\right]$

(C) Coefficient of x^{10}, in $(1-x)^{20}={}^{20}C_{10}(-1)^{10}={}^{20}C_{10}$

(D) Number of linear arrangements of 20 things of which 10 alike of one kind and rest all are different, taken all at a time $=\frac{20!}{10!}$

4.(BCD)

We know that, distribution of n distinct objects into r different boxes, if empty boxes are not allowed or in each box atleast one object is put $(n>r)$.

The number of ways

$$=r^n-{}^{r}C_1(r-1)^n+{}^{r}C_2(r-2)^n+......(-1)^{r-1}\,{}^{n}C_{r-1}1 \qquad ....(i)$$

Hence, $n=5, r=3$

$\therefore$ number of ways in which the different books be distributed in 3 persons, so each gets at least one is

$(3)^5-{}^{3}C_1(2)^5+{}^{3}C_2=243-3\times32+3=150$

(B) Number of parallelograms formed by one set of 6 parallel lines and other set of 5 parallel lines

$={}^{6}C_2\times{}^{5}C_2=15\times10=150$

(C) 5 different toys to be distributed among 3 children, so each gets at least one

$=(3)^5-{}^{3}C_1(2)^5+{}^{3}C_2$

$=243-3\times32+3=150$

(D) 3 mathematics professors are assigned five different lectures, so that each professor get at least one lecture.

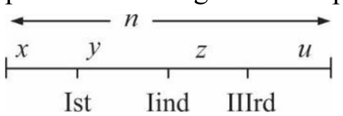
$=(3)^5-{}^{3}C_1(2)^5+{}^{3}C_2$

$=243-3\times32+3=150$

5.(BD)

Case 1 When n people are sitting a row.

Let x be the number of persons to the left of the first person chosen, y be the number of persons between the first and second, z be the number of persons between the second and the third and u be the number of persons to the right of third person.



Then, $x, u \geq 0; y, z \geq 1$ and $x + y + z + u = n - 3$

Now, Let $Now, y_1 = y - 1$ and $z_1 = z - 1$. Then, $y_1 \geq 0$ and $z_1 \geq 0$ such that $x + y_1 + z_1 + u = n - 5$, where $x, y_1, z_1, u \geq 0$.

$\therefore$ Total number of ways of selecting 3 people out of n

= Number of non-negative integral solution of $x + y_1 + z_1 + u = \text{n} - 5$

$= n - 5 = {}^{n-5+4-1}C_{4-1} = {}^{n-2}C_3$

i.e. $\quad P_n = {}^{n-2}C_3$

Case II When n people are sitting in circle .

Let n persons be denoted by $a_1, a_2, ..., a_n$.

Now, we have to select 3 persons so that no two of them are consecutive. For this we first find the number of ways in which 2 or 3 persons are consecutive. Now, number of ways in which 2 or 3 persons are consecutive is obtained as follows, with a_1 the number of such triples is

$a_1a_2a_3, a_1a_2a_5, a_1a_2a_6 .... a_1a_2a_{n-1}$.

[Note that we have excluded $a_1a_2a_n$, since it will be repeated again for, if we start with a_n, then we shall get triples : $a_n, a_na_1a_2, a_na_1a_3$]

So, number of such triples when we start with a_1 is $(n-3)$. Similarly, with $a_2, a_3, a_4, ...$ we shall get the same number of triples that is $(n-3)$.

But total number of triples is ${}^nC_3$.

Thus required number of ways $= {}^nC_3 - n(n-3)$

6.(ABD)

$$T_r = \frac{(r^2 + r - 1) \cdot r!}{(r^2 + 3r + 2)} = \frac{\left[(r+1)^2 - (r+2)\right] \cdot r!}{(r^2 + 3r + 2)} = \left[\frac{(r+1) \cdot (r+1)! - r!(r+2)}{(r+1)(r+2)}\right]$$

$$T_r = \frac{(r+1)!}{r+2} - \frac{r!}{(r+1)} \Rightarrow T_r = V_r - V_{r-1}$$

$$T_1 = V_1 - V_0$$

$$T_2 = V_2 - V_1$$

$$\vdots$$

$$T_n = V_n - V_{n-1}$$

$$S_n = T_1 + T_2 + \ldots\ldots + T_n$$

$$S_n = V_n - V_0$$

$$S_n = \frac{(n+1)!}{n+2} - \frac{1}{2}$$

$$S_n = \frac{2(n+1)! - (n+2)}{2(n+2)}.$$

7.(B)

$$az^2 + bz + c = 0 \qquad \ldots.(i)$$

$$\overline{az^2 + bz + c} = \overline{0}$$

$$\Rightarrow \quad \overline{a}\left(\overline{z}\right)^2 + \overline{b}\overline{z} + \overline{c} = 0$$

For purely imaginary root,

$$\overline{z} = -z$$

Then, $\overline{a}z^2 - \overline{b}z + \overline{c} = 0$ …(ii)

From eqs. (i) and (ii), we get

$$\frac{z^2}{b\overline{c} + \overline{b}c} = \frac{z}{c\overline{a} - \overline{c}a} = \frac{1}{-a\overline{b} - \overline{a}b}$$

$$\Rightarrow \quad z = \frac{b\overline{c} + \overline{b}c}{c\overline{a} - \overline{c}a} = \frac{c\overline{a} - \overline{c}a}{-a\overline{b} - \overline{a}b}$$

$$\therefore \quad \left(a\overline{b} + \overline{a}b\right)\left(b\overline{c} + \overline{b}c\right) + \left(c\overline{a} - \overline{c}a\right)^2 = 0$$

8.(D)

$$\because \quad az^2 + bz + c = 0$$

$$\therefore \quad \overline{az^2 + bz + c} = \overline{0}$$

$$\Rightarrow \quad \overline{a}\left(\overline{z}\right)^2 + \overline{b}\overline{z} + \overline{c} = 0$$

Since, both roots are purely imaginary.

$$\therefore \quad \overline{z} = -z$$

Then , $\overline{a}z^2 - \overline{b}z + \overline{c} = 0$

Hence, Eqs. (i) and (ii) are identical

$$\therefore \quad \frac{a}{\overline{a}} = -\frac{b}{\overline{b}} = \frac{c}{\overline{c}}$$

9.(B) If curves intersect, the parameters are equal at the point of intersection.

Hence the curves intersect where

$$a^2+bt+c_1=-at^2-bt-c_2$$

$$\Rightarrow \quad 2at^2+2bt+c_1+c_2=0 \qquad \text{....(1)}$$

The equation has distinct roots if

$$4b^2-4.2a(c_1+c_2)>0 \Rightarrow b^2>2a(c_1+c_2)$$

10.(A) For the curve S_1, we have $\dfrac{dy}{dx}=\dfrac{a}{2at+b}$

For the cure S_2, we have $\dfrac{dy}{dx}=-\dfrac{a}{2at+b}$

The curves intersect orthogonally if, at the point of intersection,

$$\frac{a}{2at+b}\times-\frac{a}{2at+b}=-1$$

$$\Rightarrow \quad 4a^2t^2+4abt+b^2-a^2=0$$

$$\Rightarrow \quad (2at+b-a)(2at+b+a)=0$$

$$\Rightarrow \quad t=\frac{a-b}{2a} \text{ or } -\frac{a+b}{2a}$$

If the curve intersect then $t=\dfrac{a-b}{2a}$ and $t=-\dfrac{a+b}{2a}$ must satisfy equation (1)

So, for $t=\dfrac{a-b}{2a}$, from (1) we get

$$\frac{2a(a-b)^2}{4a^2}+\frac{2b(a-b)}{2a}+c_1+c_2=0$$

$$\Rightarrow \quad a^2-b^2+2a(c_1+c_2)=0$$

Similarly, for $t=-\dfrac{a+b}{2a}$

we get $a^2-b^2+2a(c_1+c_2)=0$

$\therefore$ The curves intersect orthogonally if $b^2>2a(c_1+c_2)$ and $a^2-b^2+2a(c_1+c_2)=0$

11.(A) Let $C=1+\dfrac{1}{2}\cos 2\alpha-\dfrac{1}{2.4}\cos 4\alpha+\dfrac{1.3}{2.4.6}\cos 6\alpha-.........$

and $S=\dfrac{1}{2}\sin 2\alpha-\dfrac{1}{2.4}\sin 4\alpha+\dfrac{1.3}{2.4.6}\sin 6\alpha-.......$

$$\therefore \quad C+iS=1+\frac{1}{2}e^{i2\alpha}-\frac{1}{2.4}e^{i4\alpha}+\frac{1.3}{2.4.6}e^{i6\alpha}-.......$$

$$=\left(1+e^{2i\alpha}\right)^{1/2}=(1+\cos 2\alpha+i\sin 2\alpha)^{1/2}$$

$$=\left[2\cos^2\alpha+i2\sin\alpha\cos\alpha\right]^{1/2}$$

$$=\sqrt{2\cos\alpha}\left(\cos\alpha+i\sin\alpha\right)^{1/2}\quad\left[\therefore -\frac{\pi}{2}<\alpha<\frac{\pi}{2}\right]$$

$$=\sqrt{2\cos\alpha}\left(\cos\frac{\alpha}{2}+i\sin\frac{\alpha}{2}\right)$$

Equating real part

$$C=\sqrt{2\cos\alpha}\cos\frac{\alpha}{2}=\sqrt{\cos\alpha\left(1+\cos\alpha\right)}$$

12.(A) Let $C=1+n\frac{a}{c}\cos B+\frac{n(n+1)}{2!}\frac{a^2}{c^2}\cos 2B+.....$

$$S=n.\frac{a}{c}\sin B+\frac{n(n+1)}{2!}\frac{a^2}{c^2}\sin 2B+......$$

$$\therefore\qquad C+iS=1+n\frac{a}{c}e^{iB}+\frac{n(n+1)}{2!}\frac{a^2}{c^2}e^{2iB}+....$$

$$=\left(1-\frac{a}{c}e^{iB}\right)^{-n}=\left[1-\frac{\sin A}{\sin C}(\cos B+i\sin B)\right]^{-n}$$

$$=\sin^n C\left[\sin C-\sin A\cos B-i\sin A\sin B\right]^{-n}$$

$$=\sin^n C\left[\sin(B+A)-\sin A\cos B-i\sin A\sin B\right]^{-n}$$

$$=\sin^n C\left[\sin B\cos A-i\sin A\sin B\right]^{-n}$$

$$=\frac{\sin^n C}{\sin^n B}(\cos A-i\sin A)^{-n}=\frac{c^n}{b^n}(\cos nA+i\sin nA)$$

Equating imaginary parts $S=\frac{c^n}{b^n}\sin nA$

1.(250) $(1-x)^{200}={}^{200}C_0-{}^{200}C_1x+{}^{200}C_2x^2-{}^{200}C_3x^3+.........$ (i)

and $\left(\frac{1-x^{149}}{1-x}\right)=\left(x^{148}+x^{147}+x^{146}+......+x+1\right)$ ……….(ii)

On multiplying and equating coefficient of x^{148}. we get

$${}^{200}C_0-{}^{200}C_1+{}^{200}C_2.......+{}^{200}C_{148}$$

$$=\text{ coefficient of } x^{148} \text{ in } \left[(1-x)^{200}.\frac{(1-x^{149})}{(1-x)}\right]$$

$$=\text{ Coefficient of } x^{148} \text{ in } \left[(1-x)^{199}\left(1-x^{149}\right)\right]$$

$$=\text{Coefficient of } x^{148} \text{ in } (1-x)^{199}={}^{199}C_{148} \text{ or } {}^{199}C_{51}.$$

2.(511) A rational number of the desired category is of the form

$2021.x_1x_2\dots x_k$, where $1 \le k \le 9$

and $9 \ge x_1 \ge x_2 \ge \dots\dots \ge x_k \ge 1$. we can choose k digits out of 9 in ${}^9C_k$ ways and arrange them in decreasing order in just one way.

Thus, the desired number of rational numbers is

${}^9C_1 + {}^9C_2 + \dots\dots + {}^9C_9 = 2^9 - 1$

3.(1) The roots of the equation

$x = \dfrac{-n \pm \sqrt{n^2 - 4n}}{2}$ may be integers if $n^2 - 4n = I^2$ where I is an integer.

$\Rightarrow \quad n^2 - 4n - I^2 = 0 \qquad \dots (1)$

$\Rightarrow \quad n = 2 \pm \sqrt{4 + I^2}$

Now n is an integer $\Rightarrow \sqrt{4 + I^2}$ should be an integer.

$\Rightarrow \quad 4 + I^2 = k^2$ where k is an integer .

$\Rightarrow \quad k^2 - I^2 = 4$

which is possible only when $k = \pm 2,\ I = 0.$

putting $I = 0$ in (1),

$n^2 - 4n = 0 \Rightarrow n = 0,\ 4.$ For both these values,

$x^2 + nx + n = 0$ has integral roots.

$\therefore\ n = 0, 4 \Rightarrow n^2 - 4n = 0.$

4.(1)

5.(2) Here, $S = \sum_{k=1}^{\infty} \dfrac{1}{K^2} = 1^2 + \left(\dfrac{1}{2}\right)^2 + \left(\dfrac{1}{3}\right)^2 + \dots\dots +$

$$t = \sum_{k=1}^{\infty} \frac{(-1)^{K+1}}{K^2} = (1)^2 - \left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^2 - \dots\dots\dots\dots\dots\dots$$

$$S - t = 2.\left\{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{6}\right)^2 + \dots\dots\dots + \right\} = 2.\frac{S}{4} = \frac{S}{2}$$

$$\Rightarrow \quad \frac{S}{2} = t \qquad \therefore \quad \frac{S}{t} = 2$$

6.(3) $\left|k + z^2\right| = |z|^2 - k = \left|z^2\right| + |k|$

$\arg\left(z^2\right) = \arg(k)$

$\Rightarrow \quad 2\arg(z) = \pi \Rightarrow \arg(z) = \dfrac{\pi}{2}.$